Computer Science & IT



Data Structure

Tree

Lecture No. 09



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Recap of Previous Lecture



Min Heap, Max will present leaf Node

Topic — CBT- Structure $h - \left(2^h - 2^{h+1} - 1\right)$

Topic — No of leaves = $\lceil \frac{n}{2} \rceil$, # of internal $\lfloor n/2 \rfloor$

Topic — Max heap, Min Heap

Topic — Insert $O(\log_2 n)$, Construction of Heap

Topic — Delete $O(\log_2 n)$

Slide

Topics to be Covered



- Topic
- Topic
- Topic
- Topic
- Topic

Another way of construction of Heap ✓

Counting of Heap ✓

Slide



Topic : Tree

Heap adjust algorithm
Counting of heap
Graph representation
Traversal

Breadth first Search traversal

Min Heap maximum

Leaf Node any leaf Node

$\lceil \frac{n}{2} \rceil = O(n)$

Slide



Topic : Heap

Construction of Heap

1. Inserting element one after another

2. Build heap algorithm (Heapify)

Any it transform into

Max heap

Min Heap

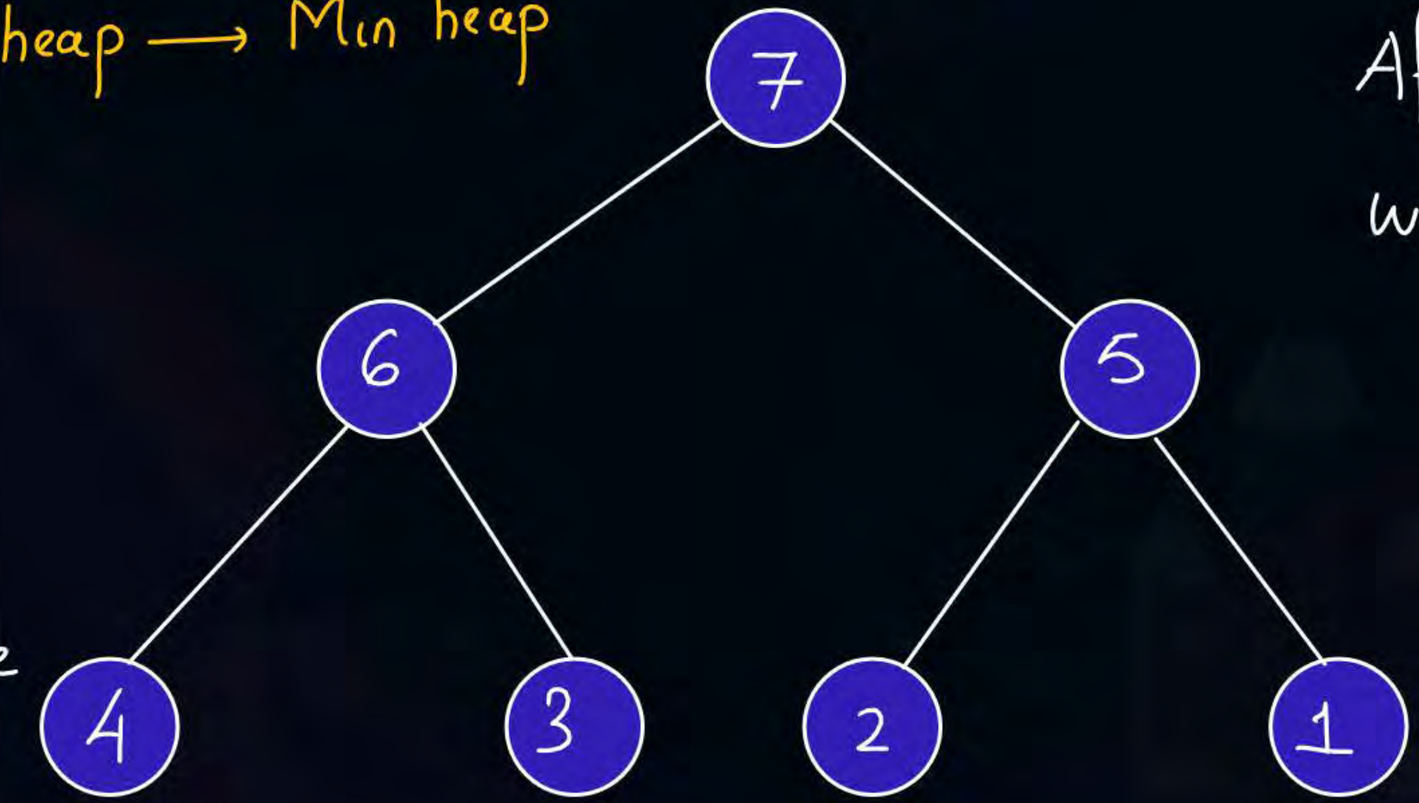
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Topic : Heap

Build heap

Max heap ⟶ Min heap



After deletion we Adjust the Root

Dont Adjust the Leaf Node ⟶

Slide



Topic : Heap

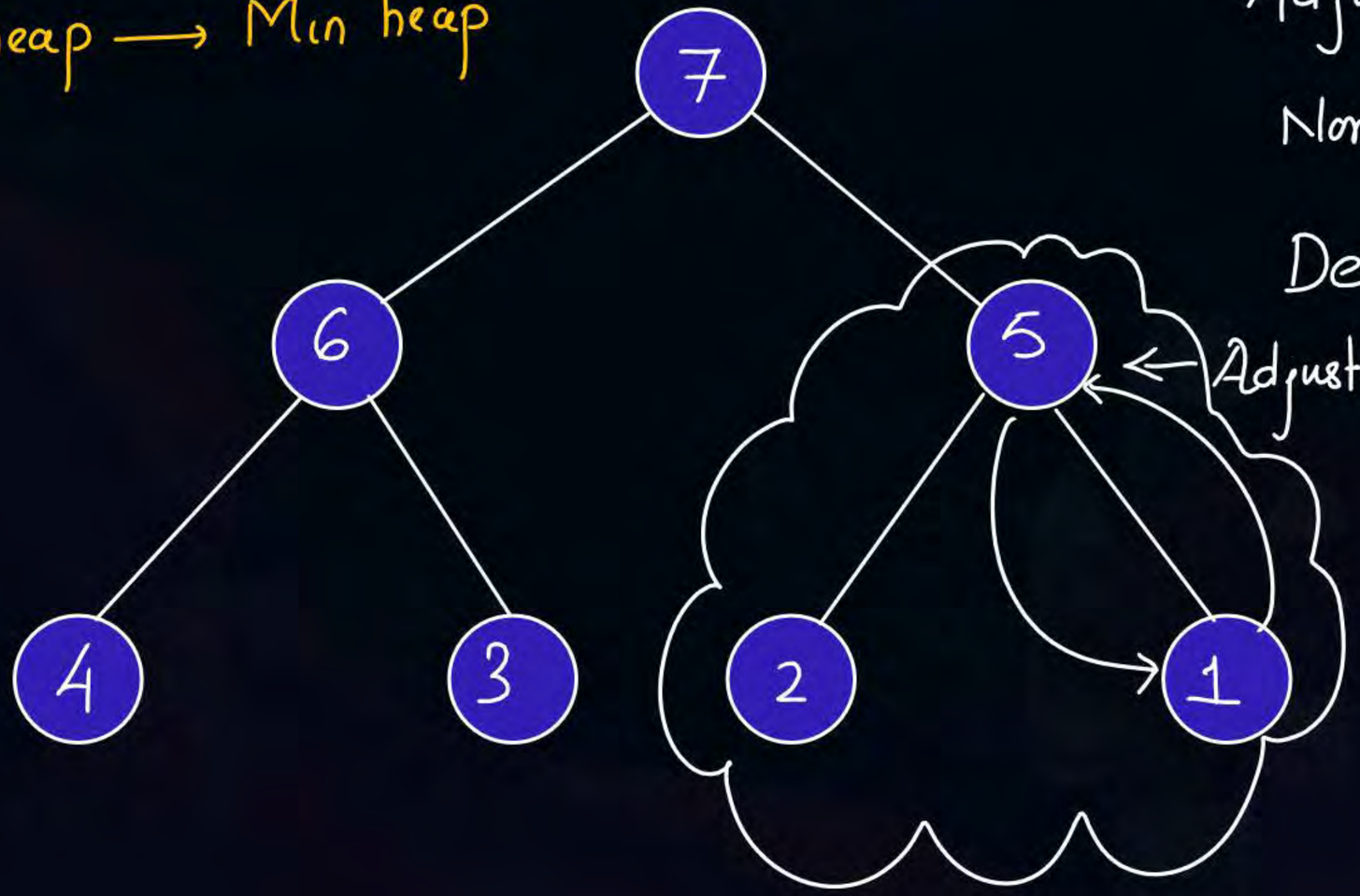
Build heap

Max heap ⟶ Min heap

Adjust

Non Leaf Node

Decremental



Slide

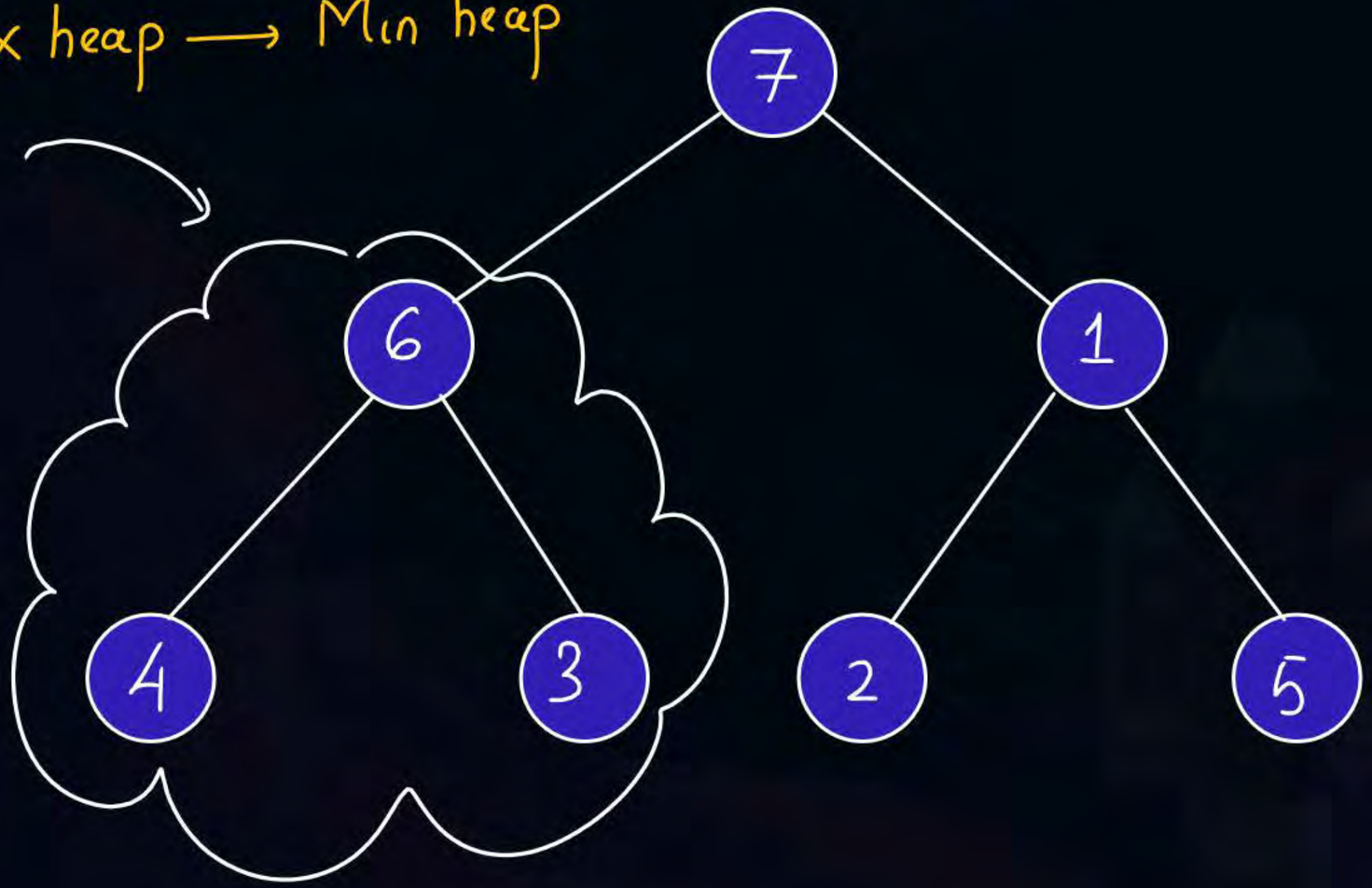


Topic : Heap

Build heap

Max heap ⟶ Min heap

Adjust 6



Slide

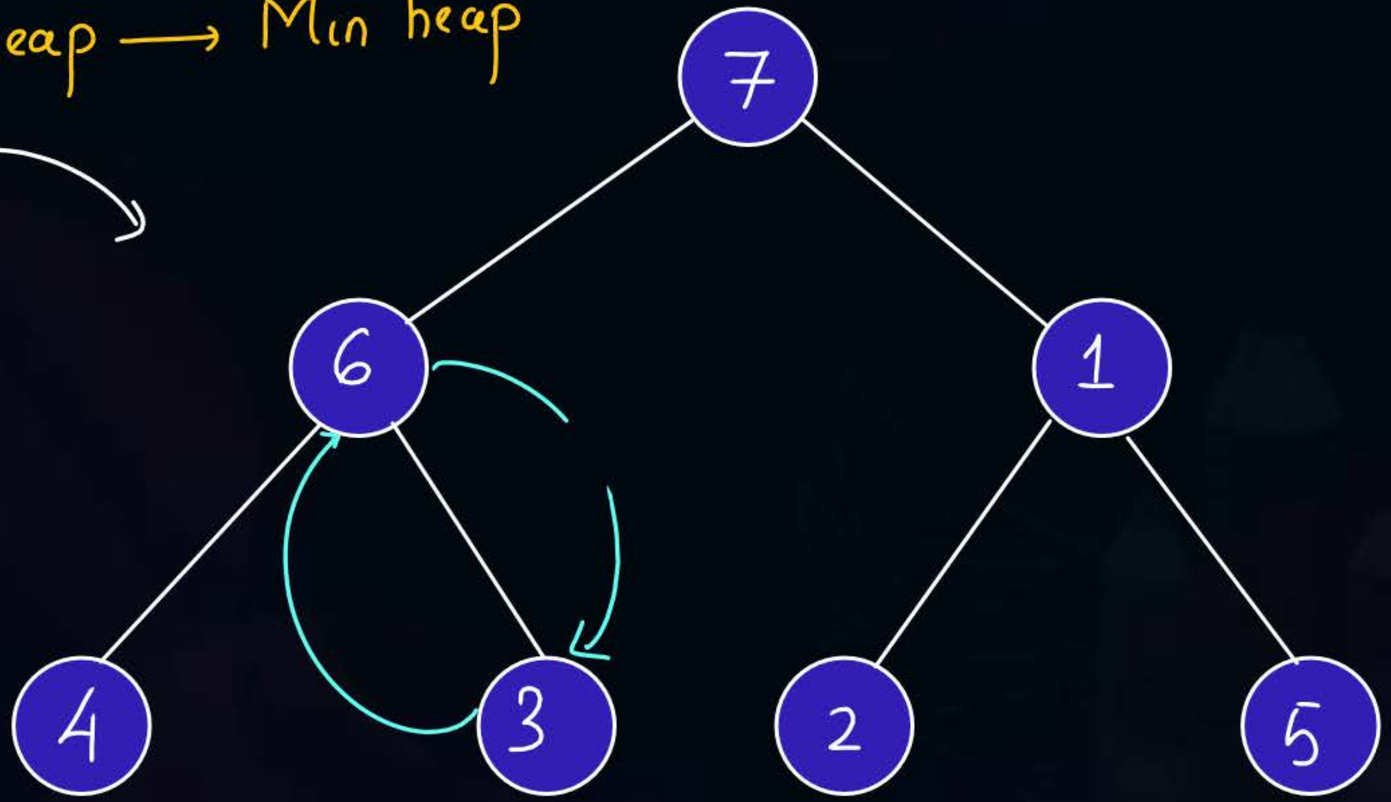


Topic : Heap

Build heap

Max heap ⟶ Min heap

Adjust 6



Slide

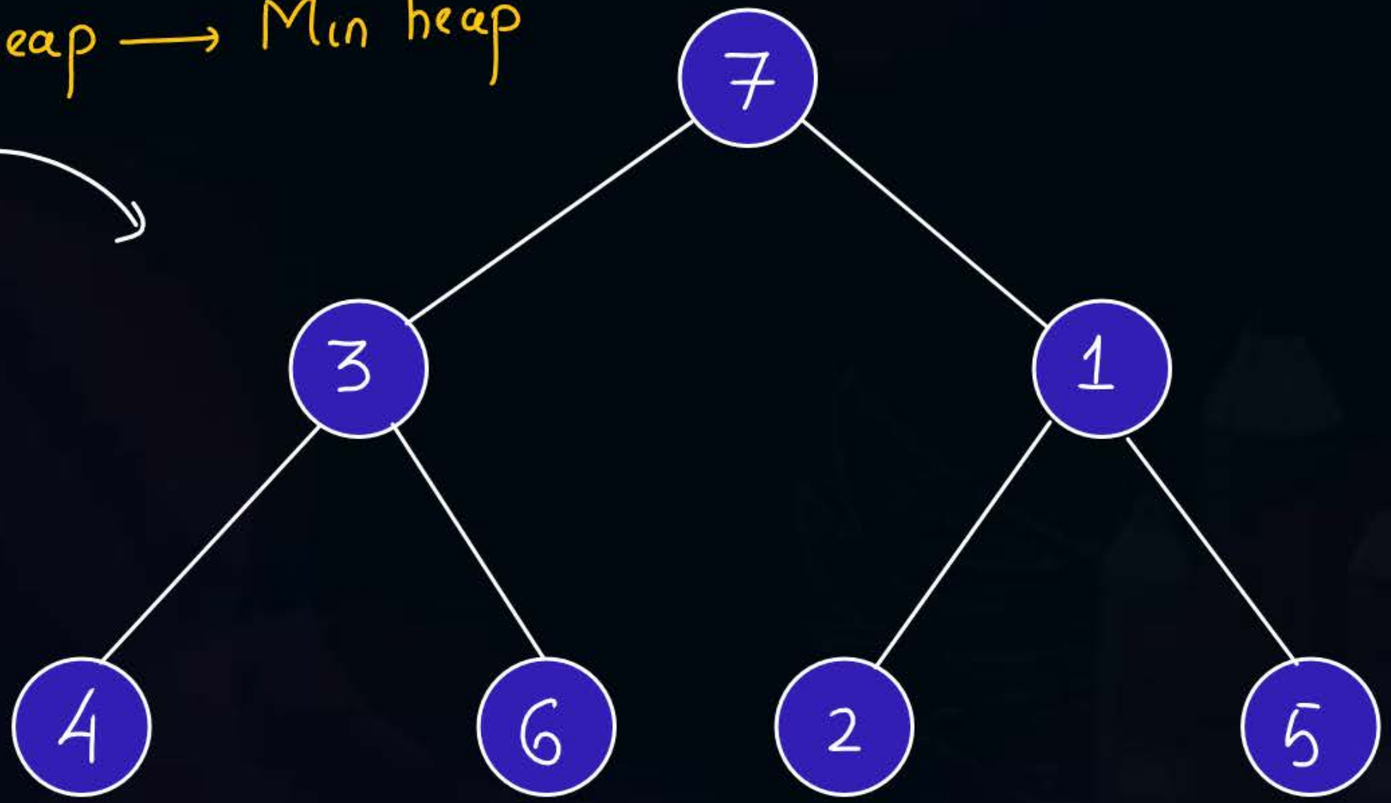


Topic : Heap

Build heap

Max heap ⟶ Min heap

Adjust 6



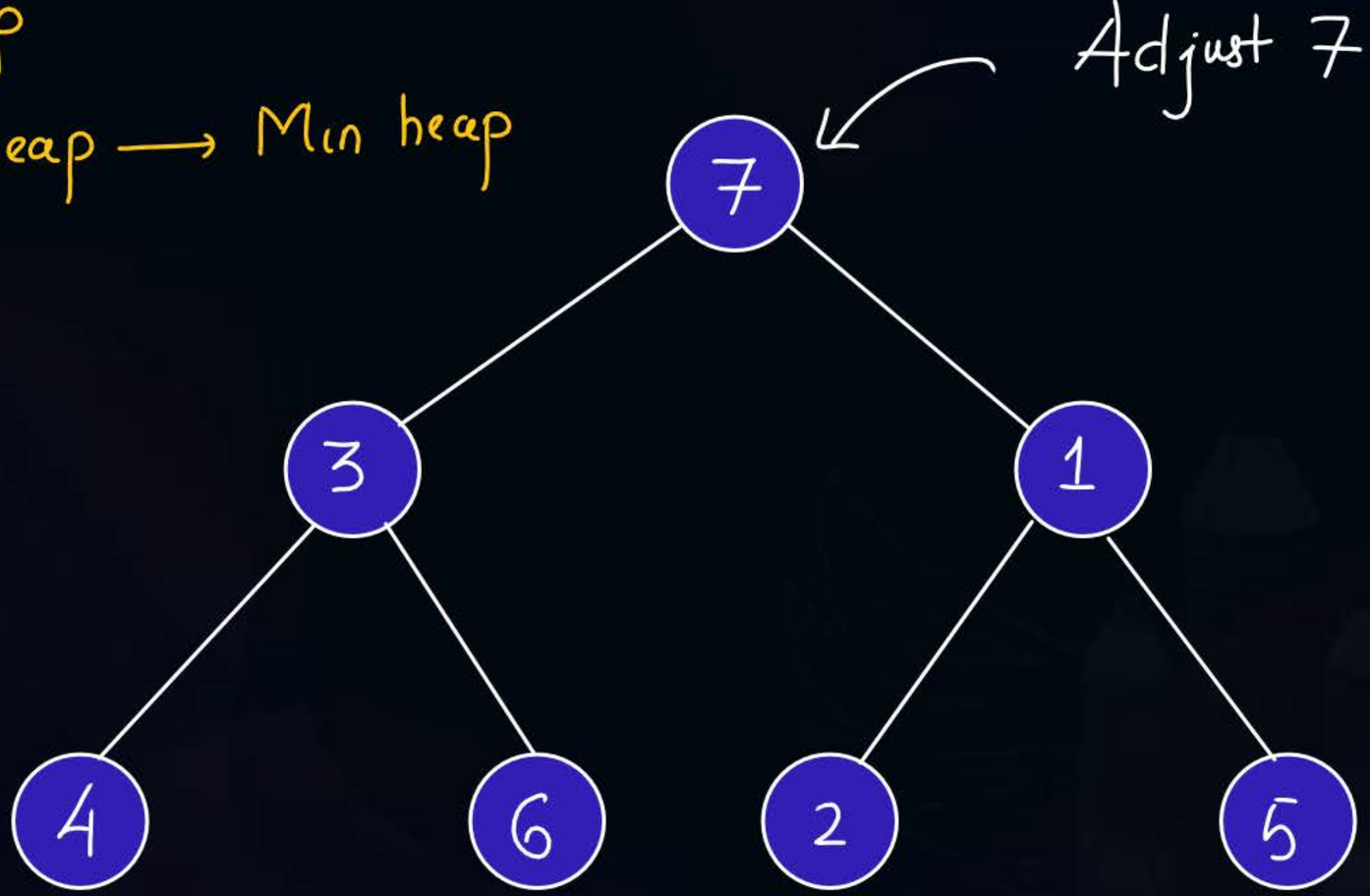
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Topic : Heap

Build heap

Max heap ⟶ Min heap



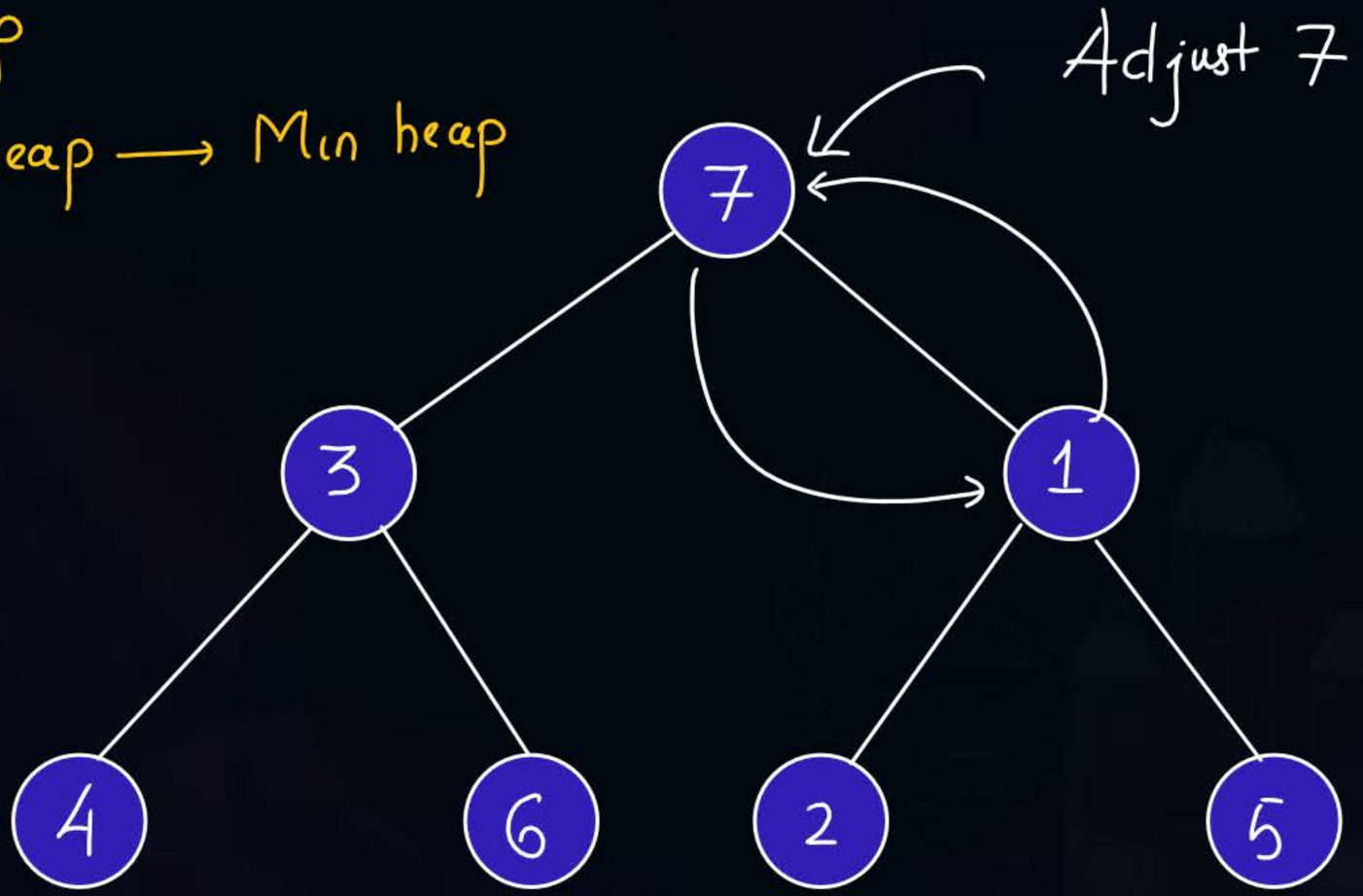
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Topic : Heap

Build heap

Max heap ⟶ Min heap



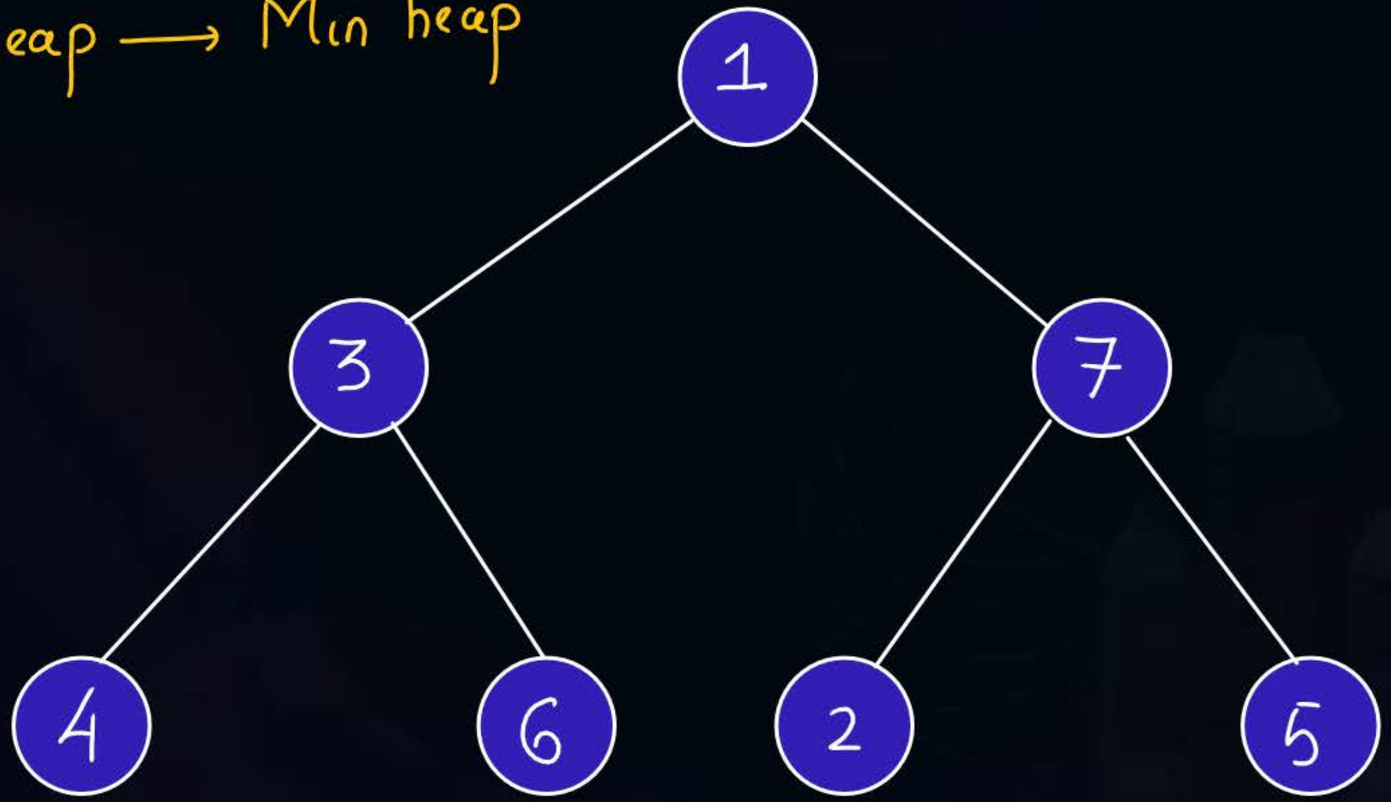
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Topic : Heap

Build heap

Max heap ⟶ Min heap



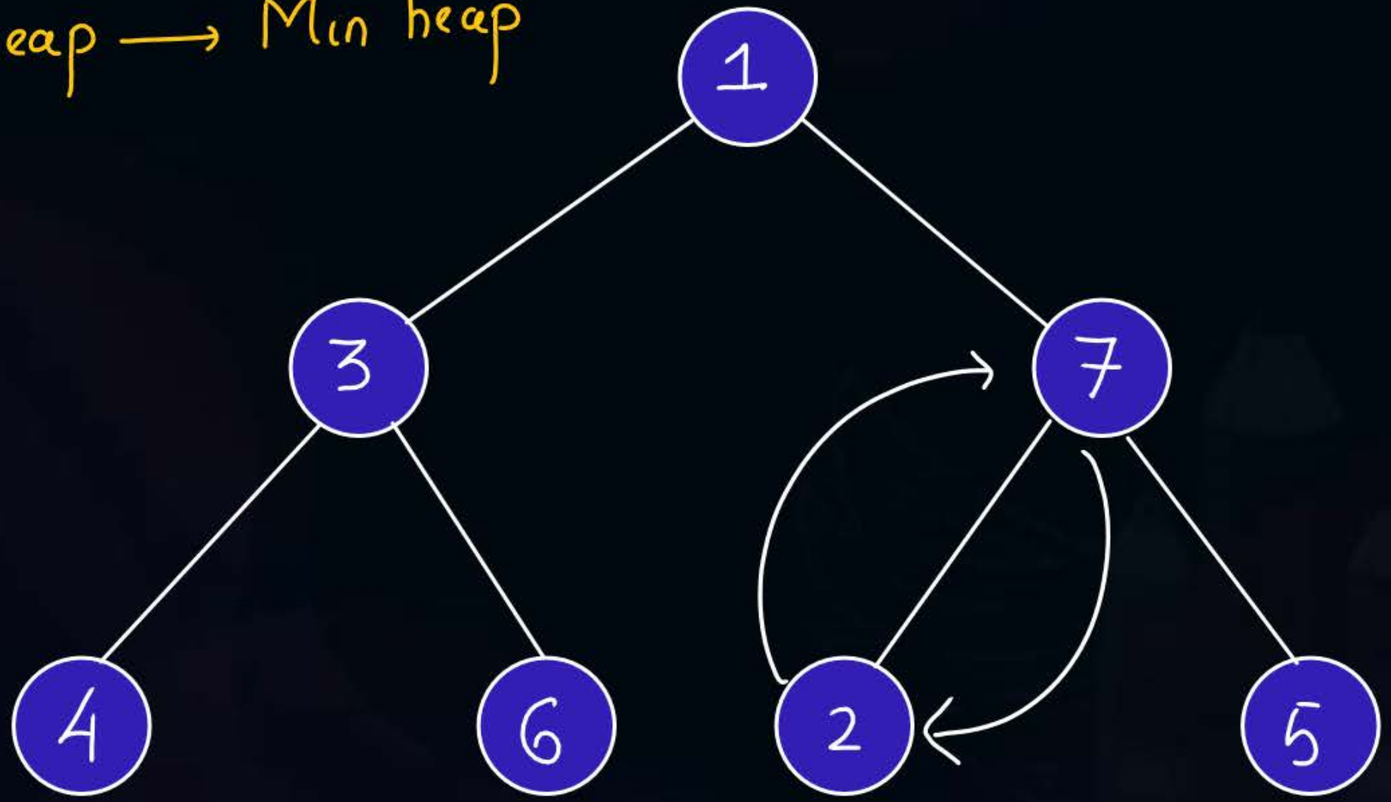
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Topic : Heap

Build heap

Max heap ⟶ Min heap



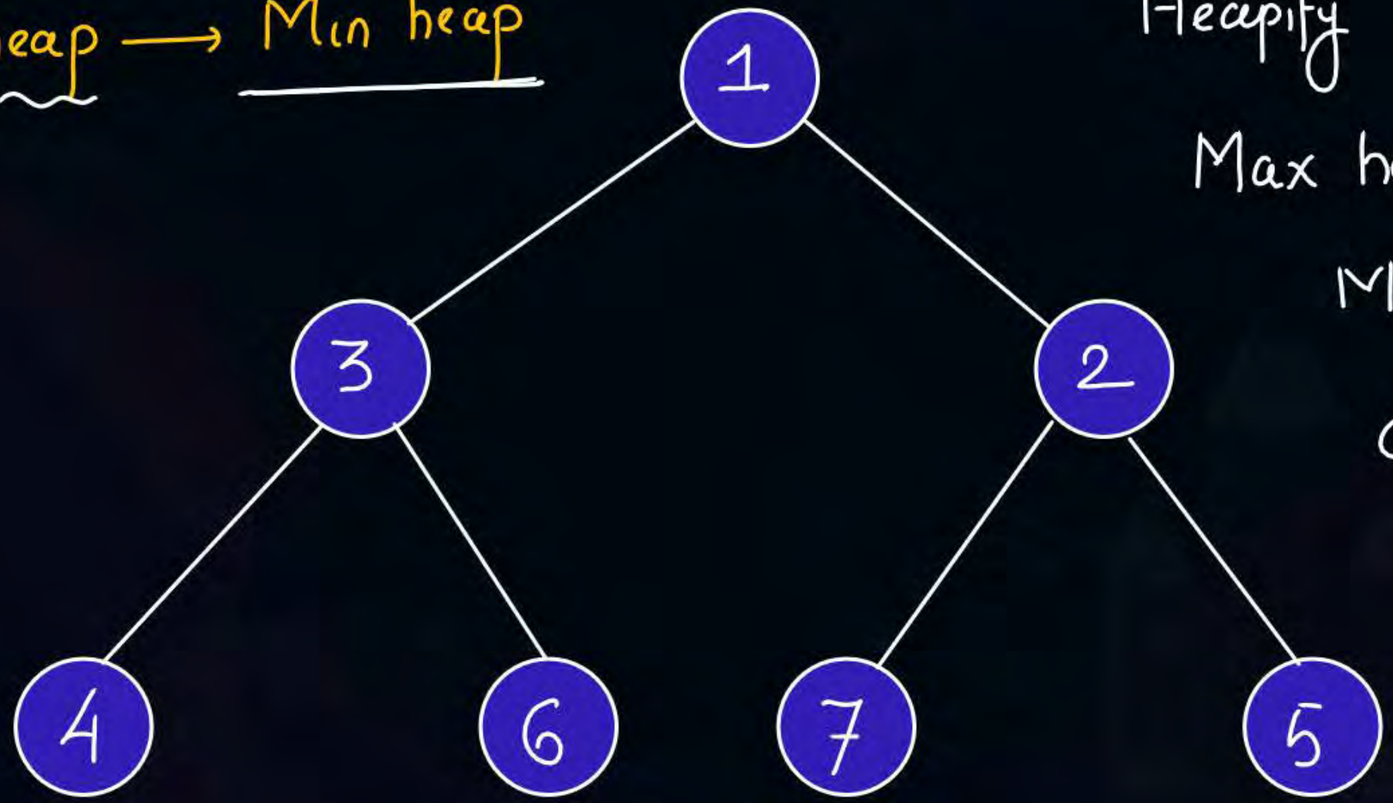
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Topic : Heap

Build heap

Max heap ⟶ Min heap



Heapify

Max heap ⟶ Min Heap

Convert

Slide



Question

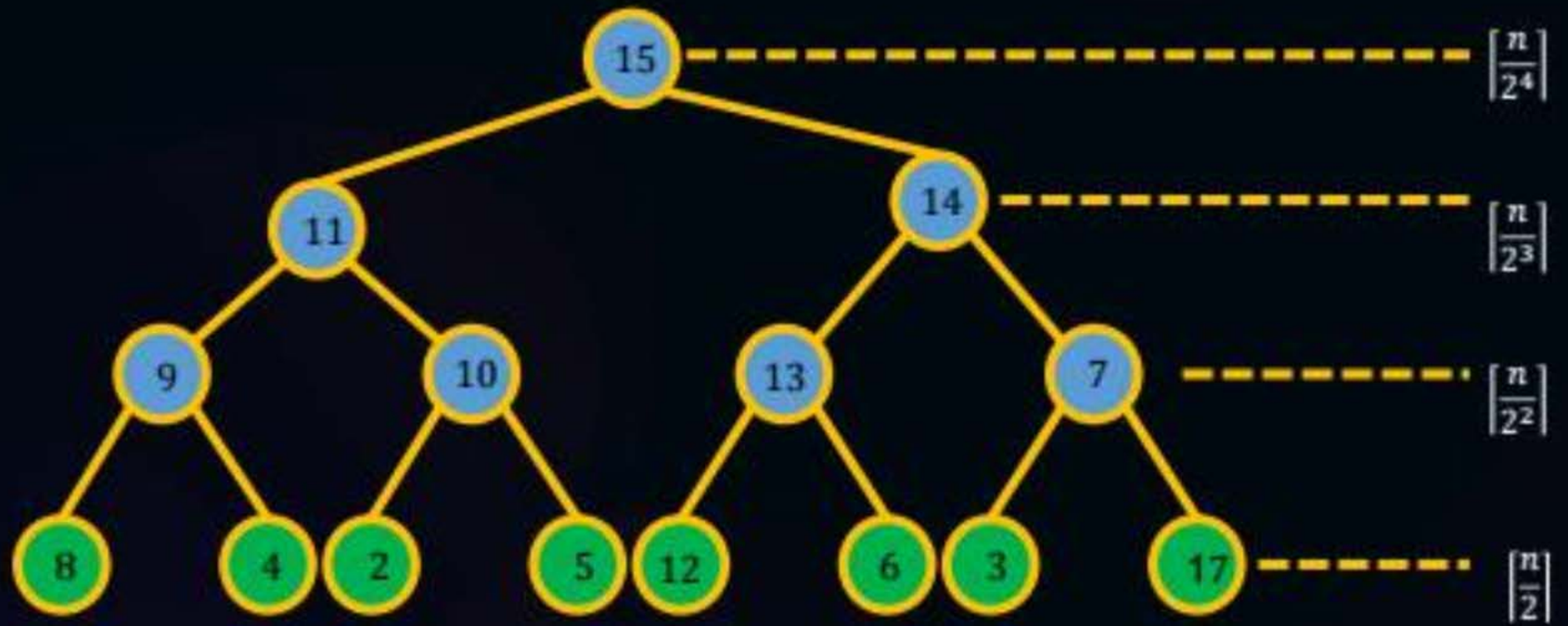
#Q. The minimum number of interchanges needed to convert the array into a max-heap is:
89, 19, 40, 17, 12, 10, 2, 5, 7, 11, 6, 9, 70

A 0

B 1

C 2

D 3



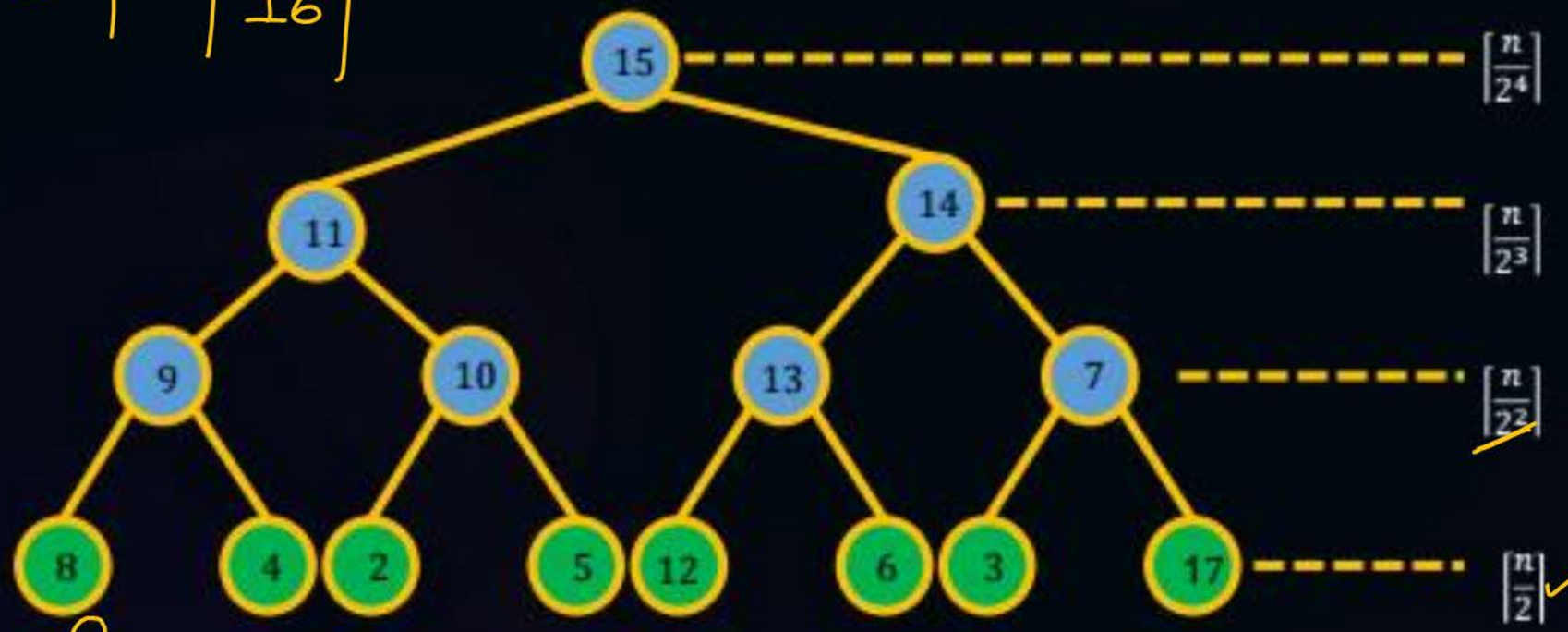
$n/2$ Internal Node ✓

$n/2$ Adjust $O(\log_2 n)$

$O(n \log n)$

$\lceil n/2^4 \rceil = \lceil \frac{15}{16} \rceil = 1$

$\lceil n/2^{h+1} \rceil$



$\lceil \frac{n}{2^3} \rceil = \lceil \frac{15}{8} \rceil = 2$

15

$\lceil n/2^{1+1} \rceil$ $\lceil \frac{15}{4} \rceil = 4$

Leaves are at heigh-0

$\lceil \frac{n}{2} \rceil$

MaxNo of swap

$$\frac{n}{2} \times 0 + \frac{n}{2^2} \times 1 + \frac{n}{2^3} \times 2 + \ldots + \frac{n}{2^{h+1}} * \log_2 n$$

$$\frac{n}{2}\times 0 + \frac{n}{2^2}\times 1 + \frac{n}{2^3}\times 2 + \cdots \frac{n}{2^{h+1}}\times \log n$$

$$\frac{n}{2}\left(\frac{1}{2}\times 1 + \frac{1}{2^2}\times 2 + \frac{1}{2^3}\times 3 + \cdots \frac{1}{2^h}\times \log n\right)$$

$\frac{n}{2}$ x constant

$O(n)$

$\frac{1}{(1-x)}$

$(1-x)^{-1}$

$-1(1-x)^{-2}(0-1)$

$(1-x)^{-2}$

$x<1$ $\sum_{i=0}^{\infty} x^i = \frac{1}{1-x}$

differeciation both side

$\sum_{i=0}^{\infty} i x^{i-1} = \frac{1}{(1-x)^2} \Rightarrow \sum_{i=0}^{\infty} i x^i = \frac{x}{(1-x)^2}$

multipy x both side

$$\sum_{i=0}^{\infty} i x^i = \frac{x}{(1-x)^2}$$

$x = 1/2$

$$= \frac{1 \times 1}{2} + \frac{2 \times 1}{2^2} + \frac{3 \times 1}{2^3} + \cdots \infty = \frac{x}{(1-x)^2}$$

Constant

$$= \frac{1/2}{(1/2)^2}$$

$$= 2$$



Topic : Heap

Counting Number of Heaps

n No. of Node

key = 11

n = 1



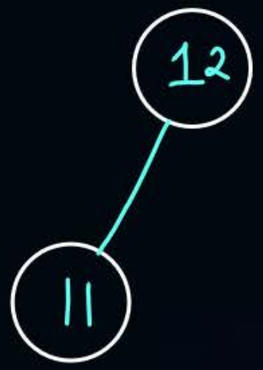
Heap counting

n = 2

key, 11, 12

Max heap

Structure CBT





Topic : Heap

Counting Number of Heaps

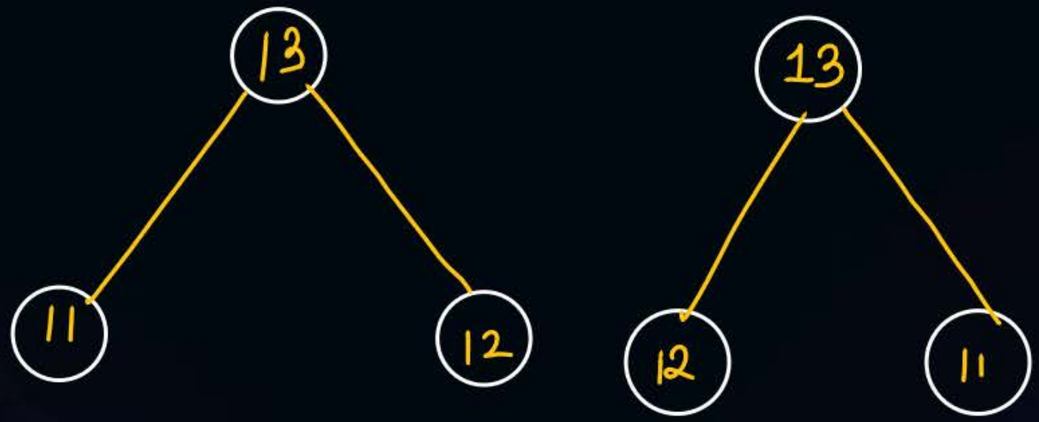
$n = 3$

key value : 11, 12, 13

Max heap :

2 max heaps

Construction of BT

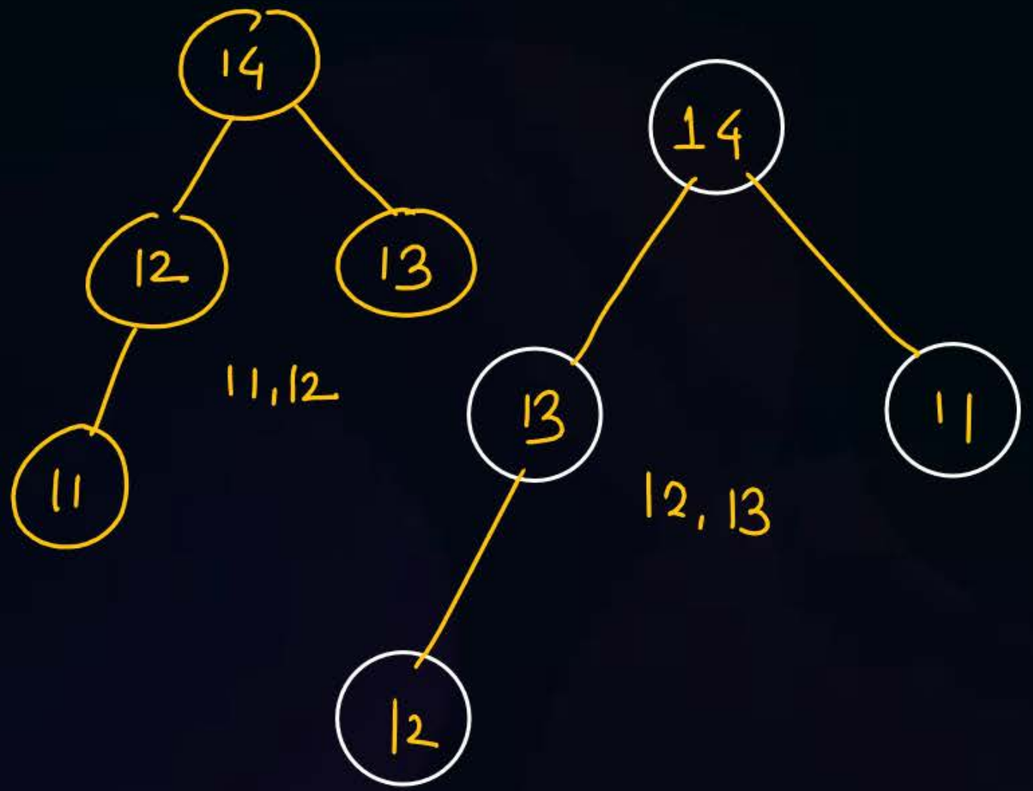


Topic : Heap

Counting Number of Heaps

$n = 4$ key 11, 12, 13, 14

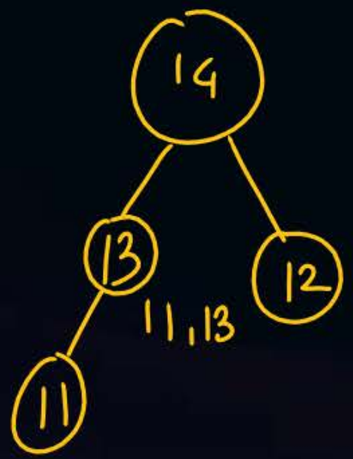
No. of max heap



Selection

out of 3 Node 2 Node to be selected

$${}^3C_2 = 3$$



Topic : Heap

Counting Number of Heaps

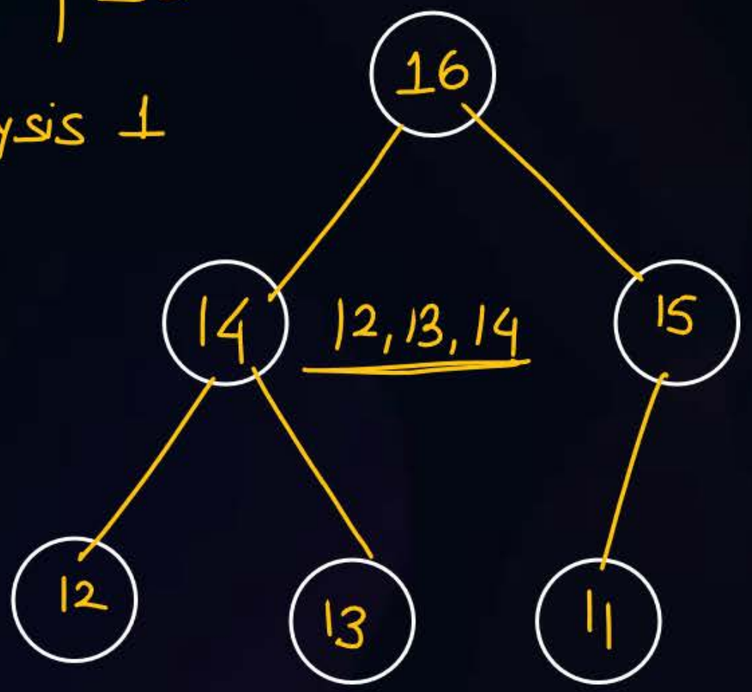
$n = 6$, 11, 12, 13, 14, 15, 16

1 is Root Rest 5

Structure of CBT

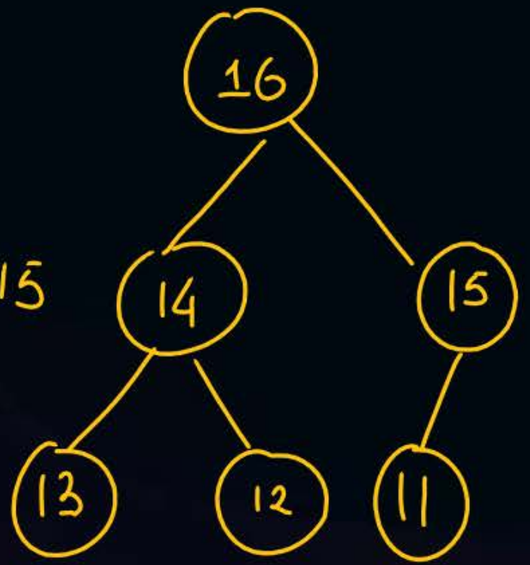
out of 10

Analysis 1



11, 15

(I) Select 3 Node out of 5 Node $^5C_3 = \frac{5}{\lfloor 2 \lfloor 3} = 10$



total

$= 10 \times 2 \times 1 = 20$



Topic : Heap

Counting Number of Heaps

$t(n)$

n elements 1 will become Root

Structure of CBT

No. of Node in left subtree

Suppose k node

Selection of left subtree

$^{n-1}C_k$

No of Heap left $t(k)$

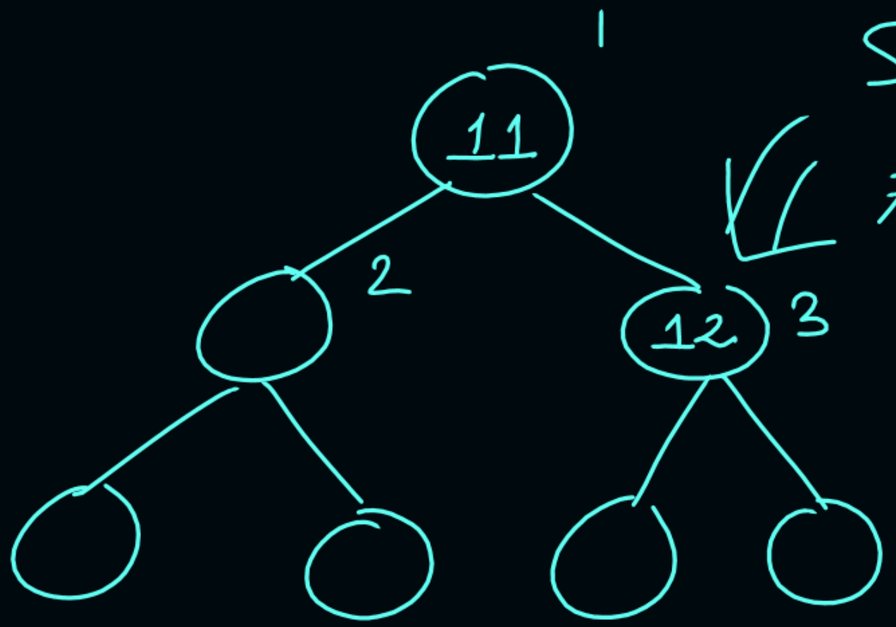
No. of heaps right subtree $t(n-1-k)$

$$t(n) = {}^{n-1}C_k \; t(k) * t(n-1-k)$$

$t(3) = 2$ $t(4) = 3$

$t(1) = 1$ $t(2) = 1$

Calculate only for small value



1 1, 12, 13, 14, 15, 16, 17

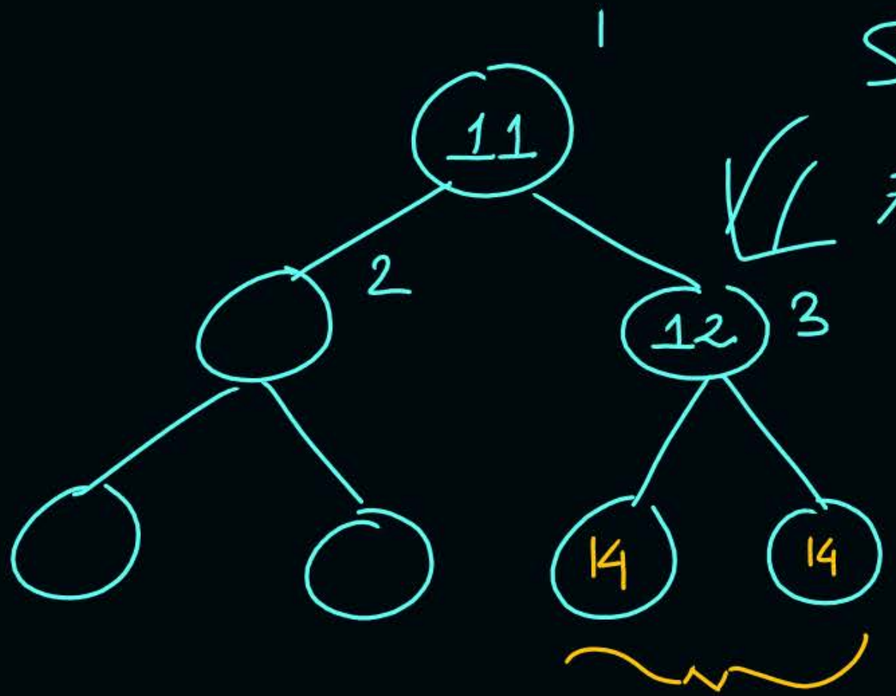
↑

Secmd Smallest element

#Q if CBT structure is given for min heap and Second Smallest element present at index 3.

What are the No. of posshon where 4th Smallest may be present

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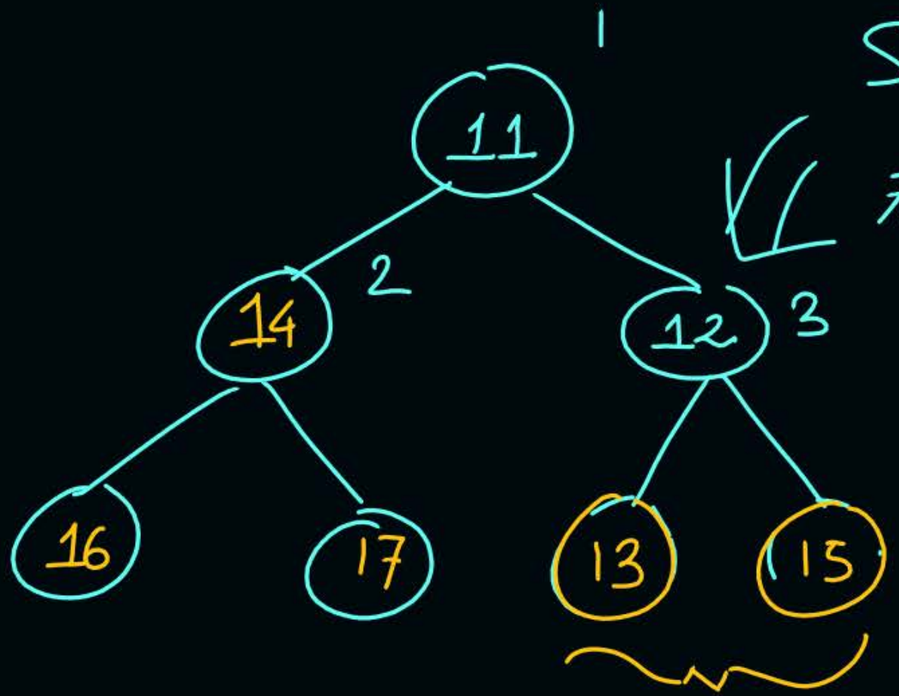


1 1, 12, 13, 14, 15, 16, 17

Second Smallest element

#Q if CBT structure is given for min heap and Second Smallest element present at index 3.

What are the No. of position where 4th Smallest may be present

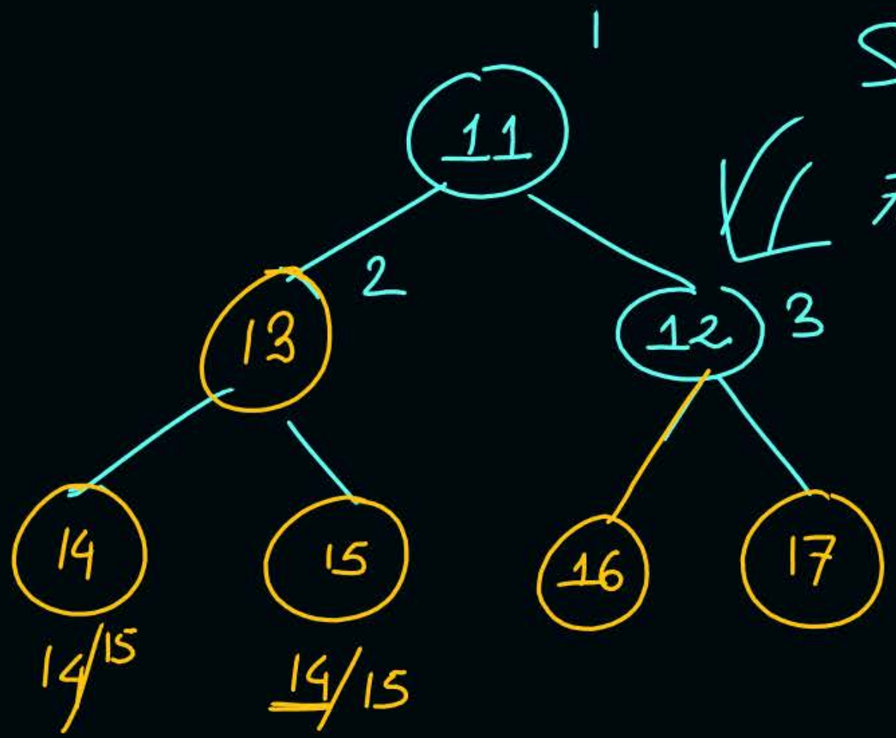


1 1, 12, 13, 14, 15, 16, 17

Second Smallest element

#Q if CBT structure is given for min heap and Second Smallest element present at index 3.

What are the No. of position where 4th Smallest may be present



1 1, 12, 13, 14, 15, 16, 17

↑

Secmd Smallest element

#Q if CBT structure is given for min heap and Second Smallest element present at index 3.

What are the No. of posthon where 4th Smallest may be present

Question: Heapify

2024

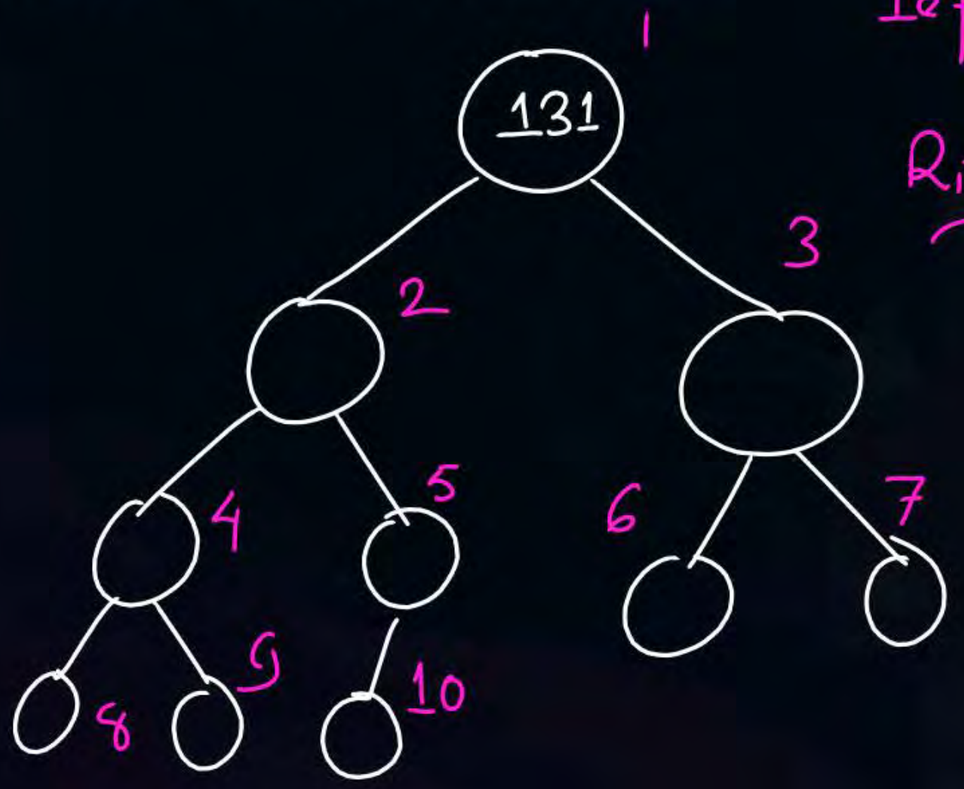
An array [82, 101, 90, 11, 111, 75, 33, 131, 44, 93] is heapified. Which one of the following options represents the first three elements in the heapified array?

(A) 82, 90, 101 X

(B) 82, 11, 93 X

(C) 131, 11, 93 X

(D) 131, 111, 90 ✓



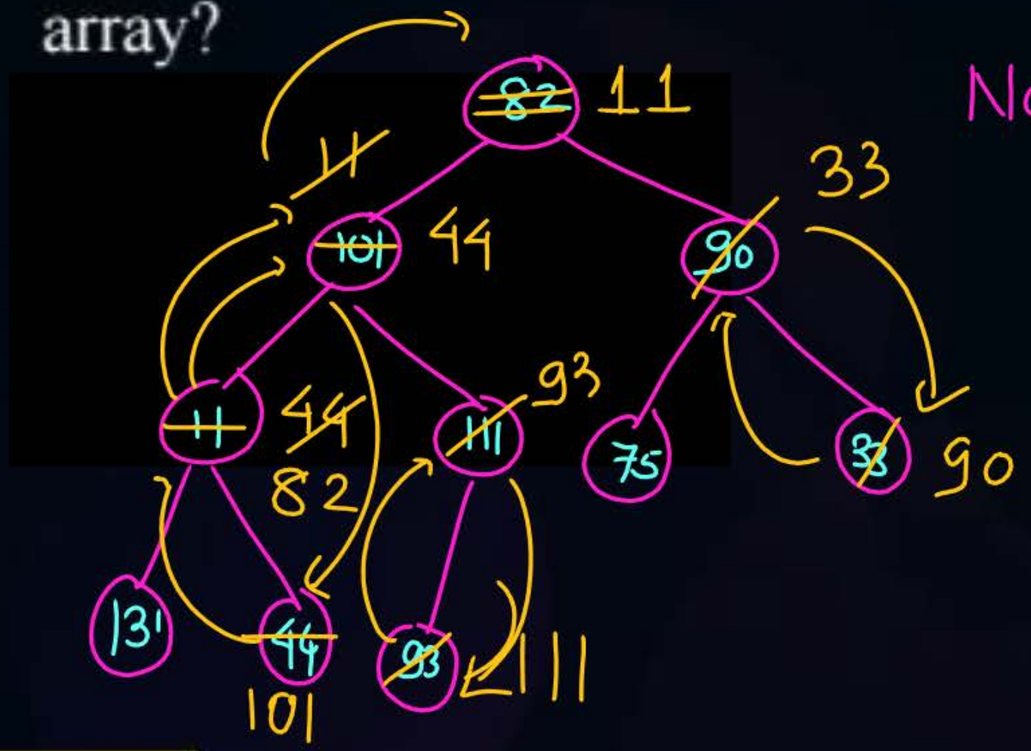
Left child

Right child

Slide

Question: Heapify

An array [82, 101, 90, 11, 111, 75, 33, 131, 44, 93] is heapified. Which one of the following options represents the first three elements in the heapified array?



① 111 & 93

② 33 & 90

③ 101 & 11, 44, 101

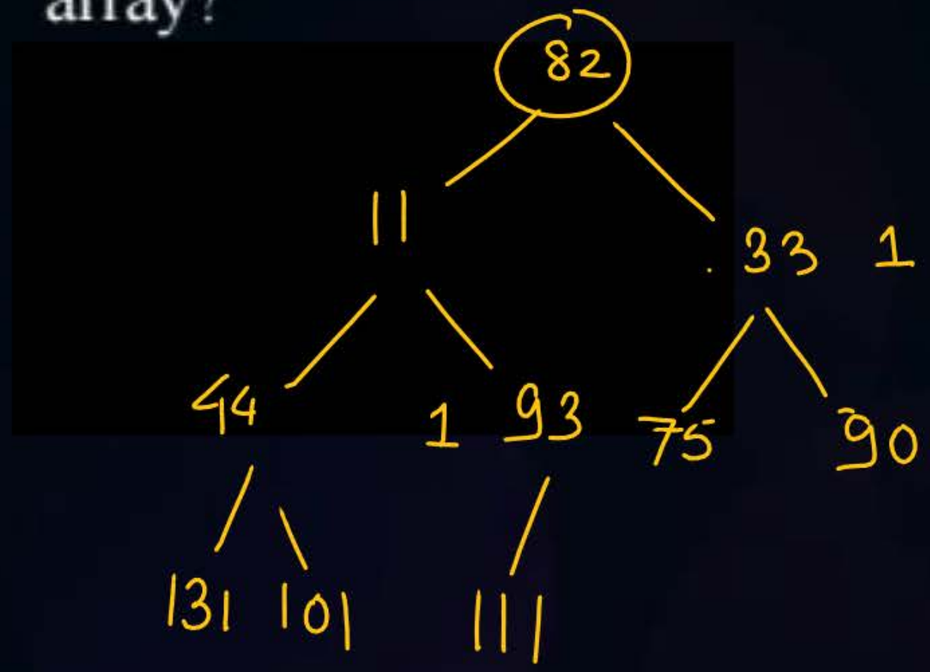
④ (11, 44) (82

No of swap required to convert the above array min heap

Slide

Question: Heapify

An array [82, 101, 90, 11, 111, 75, 33, 131, 44, 93] is heapified. Which one of the following options represents the first three elements in the heapified array?



① 111 & 93
② 33 & 90
③ 101 & 11, 44, 101
④ (11, 44) (82

No of swap required to convert the above array to min heap

Slide



2 mins Summary

Topic

Heapify Adust

Topic

Counting

Topic

Topic

Topic

Slide

THANK - YOU

Slide